

TECHNOLOGY ENTREPRENEURSHIP
ENTREPRENEUR BUSINESS IDEA



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PROBLEM

Parking on campus is often full in one spot but empty in another. Parking attendants often misestimate the availability of spaces due to manual monitoring. As a result, students often wander around looking for a spot, wasting time.

BUSINESS IDEA

- **Name & Business Concept**

ParkSmart Vision – A campus parking system based on CCTV cameras and a CNN (Convolutional Neural Network) to automatically detect empty and occupied parking spaces.

- Cameras record the parking area, and then a CNN model classifies parking spaces (empty/occupied).
- Real-time information is displayed on a mobile app and on the officer's dashboard.
- Without IoT sensors in each space, it's more cost-effective and easier to implement.

- **Target Users & Market**

- **Students & Lecturers:** Get faster information on available slots via the app.
- **Parking attendants:** Can monitor parking spaces via the dashboard without manual patrols.
- **Campus management:** Obtains parking density data for evaluation and planning.
- **Potential market:** Universities, large schools, government agencies, and companies with large parking areas.

- **Main Features**

- **Real-time Slot Detection:** The app displays the number of empty slots based on cameras and CNN.
- **Digital Parking Map:** Students/faculty can see empty parking locations before entering.
- **Smart Notification:** The system notifies you when an area is full and recommends alternative locations.
- **Usage Analytics:** Hourly/daily/weekly parking congestion reports for campus management.
- **Officer Dashboard:** Monitor all parking areas from one screen.

- **Potential Social & Economic Impacts**

- **Social:**

- **Reduces parking time:** more efficient and reduces user stress.
 - Makes campus traffic flow more orderly and smoother.
 - Transparency in parking attendants' work (more accurate, not just estimates).

- **Economic:**

- Saves operational costs by eliminating the need for IoT sensors in every parking space.
 - **Potential for data monetization:** campuses can use parking analytics for facility planning.
 - Can be developed as a SaaS service for other campuses/institutions.

- **Business Model Used**

B2B Subscription (Campus/Institution) - The campus pays the system setup fee + annual license fee.

VISUALIZATION

